

Special relativity demonstration

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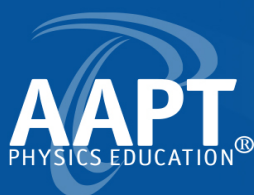
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appropriate gauge⁴ $(\mu_0/4\pi)\int(\text{curl } \mathbf{H}_2)/r_1 dV = \mathbf{A}_{12}$, hence

$$\mu_0\epsilon_0\int\mathbf{E}_1\times\mathbf{H}_2dV=q_1\mathbf{A}_{12}. \quad (2)$$

Here $\mathbf{A}_{12}$ is the vector potential at q_1 due to the motion of q_2 . The second part of the integral in Eq. (1) is similarly $q_2\mathbf{A}_{21}$, and the total canonical momentum is therefore given by

$$m_1\mathbf{v}_1+m_2\mathbf{v}_2+q_1\mathbf{A}_{12}+q_2\mathbf{A}_{21}=\mathbf{p}_{\text{tot}}+\int\mathbf{G}_{\text{int}}dV, \quad (3)$$

which is a conserved quantity, $\mathbf{p}_{\text{tot}}$ being the total particle momentum. A similar relation exists with respect to angular momentum. Generalization of the result to include any number of free charges is straightforward.

Special relativity demonstration

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"The influence of the crucial Michelson-Morley experiment upon my own efforts has been rather indirect. I learned of it through H. A. Lorentz's decisive investigations of the electrodynamics of moving bodies (1895) with which I was acquainted before developing the special theory of relativity . . . What led me more or less directly to the special theory of relativity was the conviction that the electromotive force acting on a body moving in a magnetic field was nothing else but an electric field."

Albert Einstein¹

Although many physics instructors are aware that relativity is involved in the Faraday magnet and coil induction experiment, few exploit the full significance of it as a relativity demonstration. This note is to remind teachers that this experiment demonstrates a truly relativistic effect at



Fig. 1. Moving coil case.

¹E. Breitenberger, Am. J. Phys. **36**, 505 (1968).

²C. G. Darwin, Philos. Mag. **30**, 537 (1920).

³L. Landau and E. Lifshitz, *The Classical Theory of Fields* (Addison-Wesley, Reading, MA, 1951).

⁴J. D. Jackson, *Classical Electrodynamics* (Wiley, New York, 1962).

⁵A. Shadowitz, *The Electromagnetic Field* (McGraw-Hill, New York, 1975).

⁶The relationship between field momentum and the vector potential in general has been discussed by G. T. Trammel, Phys. Rev. **134**, B1183 (1964) and by E. J. Konopinski, Am. J. Phys. **46**, 499 (1978). However, neither of these authors addressed the specific question of momentum conservation between two particles.

⁷The apparently divergent self-momentum terms involving $\mathbf{E}_1\times\mathbf{H}_1$ and $\mathbf{E}_2\times\mathbf{H}_2$ are ignored. This essentially amounts to a mass renormalization. A discussion of the self-momentum may be found in Ref. 4.

⁸This approximation is accurate to order $1/c^2$ as discussed by Breitenberger in Ref. 1.

very low velocities. The experiment shows, among other things, that physical results depend only on the relative motion (Einstein's first postulate of special relativity, the physics is independent of the uniform motion of the inertial frame), and that electric and magnetic fields manifest themselves differently to different moving observers.² In addition the experiment has the advantage of motivating special relativity in the same way as Einstein himself was motivated, as in the quote above. This demonstration is useful as a general introduction to relativity in the noncalculus courses, and as a motivation for the Lorentz transformation in a higher level special relativity course.

A good place to introduce the demonstration for the first time in a general course is just before the coverage of Faraday induction. (Later, when relativity is covered, the demonstration can be repeated and new features emphasized.) Referring to Fig. 1, first hold the magnet stationary with respect to the classroom, and move the coil toward the magnet. In this situation, case A, the force that moves the electrons around the coil to produce the galvanometer reading is the Lorentz magnetic force $\mathbf{F} = q\mathbf{v}/c \times \mathbf{B}$, where $\mathbf{v}$ is the velocity of the coil toward the magnet (see Fig. 2).

Before demonstrating the moving magnet, case B, discuss with the students what they should *expect* to see. There is a magnetic field, but it does not affect the electrons in the coil since their velocity is initially zero. Even after they begin to circulate around the coil, the magnetic $\mathbf{v} \times \mathbf{B}$ force is

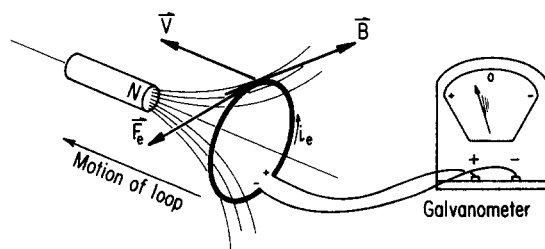


Fig. 2. Analysis of the moving coil case.



We see Lucy and Ringo both moving, approaching each other.

Lucy says:
The loop is stationary and the magnet is moving toward it. There is a magnetic field, but it can't produce any force on my electrons since they are stationary within the loop. Instead, the magnetic field is changing, growing stronger as the magnet gets closer, and this changing magnetic field produces an electric field which causes forces on the electrons, and drives them around the loop and produces the current in the galvanometer.

Ringo says:
The magnet is stationary and the loop is moving toward it. The electrons in the loop, since they are moving with the loop, feel a magnetic force,

$$\vec{F}_m = -\frac{e}{c} \vec{v} \times \vec{B},$$
which drives them around the loop and produces the current in the galvanometer.
There is no electric field.

Conclusion:
Electric and magnetic fields are not invariant entities themselves, but are aspects of a single entity, the electromagnetic field, which manifests itself differently to different moving observers.

Fig. 3. Coil and magnet relativity for noncalculus students.

perpendicular to the wire and so does not cause the current. In fact, nothing the students have studied so far in E & M would lead them to expect a galvanometer deflection in case B. When this proposition is put to the class, some students will object that "it shouldn't matter whether the coil or the magnet is moved." If the class is pressed on this point, one can usually draw out the comment that "only the relative motion should matter." After a short discussion of Einstein's relativity principle, one can go on to perform case B. But, of course, there still must be a specific physical origin for the force on the electrons, and it is then appropriate to introduce the Faraday induction or $\nabla \times \mathbf{E} = c^{-1} d\mathbf{B}/dt$ law. (In a noncalculus course this can be simply worded as a *changing* magnetic field produces a circular electric

field.)

Further points that can be emphasized when the experiment is demonstrated later for relativity are as follows:

One, it is the *relative* motion that produces the same physical result in both cases A and B, a certain deviation of the galvanometer needle.

Two, in case A there is only a magnetic field in the classroom, but in case B there is also an electric field in the classroom. It is this electric field that is moving the electrons to cause the galvanometer deflection in case B. In the older texts the force due to the electric field was called the electromotive force as in Einstein's quote above.

Three, the demonstration shows the Lorentz transformation of the electromagnetic field in that in a frame of reference moving through a stationary $\mathbf{B}$ field, there is also an $\mathbf{E}$ field.

Four, the galvanometer deflection is truly a relativistic effect, proportional to v/c . This is most easily seen in the cgs system where the current $i = (lB/R)(v/c)$, with l the length of the wire in the coil and R its resistance. It is interesting to ask the advanced students why a relativistic effect can easily be seen here even though $v/c \sim 10^{-8}$. The reason is the same reason why a current carrying wire produces a measureable magnetic field (also a relativistic effect). The magnetic field of a single electron moving with velocity $\mathbf{v}$ is

$$\mathbf{B} = -\frac{e}{r^3} \frac{\mathbf{v}}{c} \times \mathbf{r}.$$

Here the velocity is the drift velocity of the electrons $v = v_d \approx 10^{-2}$ cm/sec, so $v/c \approx 10^{-12}$, but the wire has *lots* of electrons.

The principles of the demonstration as presented in the noncalculus course are summarized in Fig. 3.

¹ Albert Einstein at a 1952 meeting honoring the centenary of Michelson's birth as quoted by Robert Resnick, *Introduction to Special Relativity* (Wiley, New York, 1968).

² This demonstration was inspired by Kenneth W. Ford, *Basic Physics* (Blaisdell, Waltham, MA, 1968), Sec. 16.7.

The history of electromagnetic induction

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There are two points I would like to bring to attention concerning the history of the search for electromagnetic induction.

First, in studies of the history of electromagnetism it is commonly assumed that experimenters in the 1820s correctly sought to discover electromagnetic induction by attempting to perform experiments opposite to those which displayed the phenomenon of electromagnetism. This search eventually met with success in 1831 with the discoveries made by Faraday. However it would appear that during the period separating the discoveries of Oersted and Faraday some experimenters based their search on an illogical deduction. Further, the illogical deduction appears not to have been noted by historians of science.

Oersted discovered that if a constant direct current flowed through a wire, then a constant magnetic field was produced around the wire. This led to experiments to obtain

the direct opposite, i.e., to produce a steady current from a steady magnetic field—a phenomenon which it was later realized is impossible to obtain.

In the frenzy of activity which followed the publication of Oersted's discovery Arago in Paris, Seebeck and Pohl in Berlin, and Davy in London, independently discovered that a current-carrying coil could magnetize iron or steel needles placed inside it. It was in seeking the converse of this experiment, probably first attempted by Fresnel, that the error of logic was made.

Apparently the converse of this was taken to be essentially the same as that given above, namely that a magnet inside a coil should induce a current in the coil. This, however, is not the converse and it is not, as reviewers have taken it to be, a logical deduction to make.

Since an unmagnetized bar became magnetized, i.e., its magnetic state was changed, by the effects of a constant